Dataset Proposal - Malaria & Dengue Early Warning (Public health track)

HackInsight 2026 · Data-to-Decision · generated from manifest.csv and the warehouse

Problem and primary user

District health officers in Sindh, Balochistan and Khyber Pakhtunkhwa decide each week where to send RDT kits, antimalarials, beds and fumigation teams. We forecast next week's suspected malaria and dengue cases per district, with an 80% range, and give cited NIH/WHO guidance in English and Urdu.

Credible sources used for model training and evaluation

Source	Publisher / URL	Licence / terms	Update	Files / MB in manifest
IDSR weekly Public Health Bulletins (PDF)	National Institute of Health, Islamabad - nih.org.pk/phb/weekly-bulletin	Public government report; robots.txt 'Disallow:' empty (all allowed); attribution	Weekly	216 / 489.2
Historical (ERA5) + forecast hourly weather	Open-Meteo - archive-api.open-meteo.com, api.open-meteo.com	CC BY 4.0; free tier non-commercial (<10k calls/day)	Hourly	32 / 73.8
OpenDengue v1.3 temporal extract	Clarke et al. 2024 - doi:10.6084/m9.figshare.24259573	CC BY 4.0	Versioned releases	1 / 52.3
Admin boundaries (ADM2 pcodes, centroids), 2017 census population, MPI	OCHA FISS / PBS / OPHI via data.humdata.org	CC BY-IGO; CC0	Rare	3 / 27.8
Advisories, guidelines, fact sheets (RAG only)	NIH Pakistan; WHO	Public; attribution	Ad hoc	7 / 8.2

Source not on the track list, and why it is credible

The NIH IDSR bulletins are the national Integrated Disease Surveillance and Response system of Pakistan's federal public-health institute, compiled from district reports. The original publisher is named on every page, files are fetched from nih.org.pk directly, and each file's SHA-256 is recorded. They are independent of Open-Meteo (weather) and of OpenDengue (literature compilation).

Volume (gathered during the event)

IDSR: 203,895 parsed cells from 177 bulletins (weeks 2021-07-19 to 2026-09-21). Weather: 2,905,770 hourly district rows (2021-01-01 to 2026-10-20). OpenDengue Pakistan: 922 rows. Gold table: 34,920 district-disease-weeks for 97 districts, 15,876 with a reported count. Manifest: 259 raw files.

Why this data fits the problem and the user

It is the same surveillance stream the health departments act on, at the district level where the allocation decision is made, combined with the weather drivers of mosquito breeding.

Declared baseline (fixed before results)

Next week's suspected cases in a district = this week's reported cases (naive persistence). Metric: mean absolute error on a time-based test period (target weeks from 2025-10-01), scored once. 'Better' = lower MAE than the baseline, with a block-bootstrap 95% interval.

Known limits and risks

Suspected not confirmed cases; each bulletin lists only each province's most frequent diseases ('not listed' kept missing, not zero); Punjab/ICT districts are 'NR'; 2021-22 bulletins use a different layout and are flagged, not guessed; no sex/age breakdown; 1-2 week publication lag; 2017 population.